



Existing factory cameras record risk.
KantaForge turns it into actionable safety evidence.

A privacy-preserving, AI-assisted workplace-safety monitoring and evidence platform for Malaysian legacy manufacturers.



THE PROBLEM :
Fragmented, reactive safety data



INITIAL CUSTOMER :
Malaysian CNC and metal-manufacturing SMEs



THE SOLUTION :
AI detection → Human review → Traceable records



REVENUE MODEL :
Factory SaaS (Setup fee + monthly monitored-zone subscription)



Factories Have Cameras. They Lack Actionable Safety Data.

How Safety Video Works Today :



CCTV records



Footage accumulates



Limited Human Review



Post-Incident Search



Separate Documentation

10,335

INJURIES IN 2023

Occupational injuries were recorded in Malaysia's manufacturing sector in 2023.

(representing ~ 26.5% of the national total)



Source : Department of Statistics Malaysia , 2023

01 Reactive, Not Preventive :

CCTV footage is reviewed after incidents, not used for continuous alerts.

02 Raw Video ≠ Evidence :

Footage lacks verified event types, locations, and resolution status.

03 Fragmented Records :

CCTV, inspection notes, and corrective-action spreadsheets remain disconnected.

Validating the **Review-Capacity Gap on the Ground**

Our CCTV is mainly reviewed after an incident because we simply cannot continuously monitor every camera.

– Mr. Tan, EHS Officer, Manufacturing Company, Interviewed on 5 August 2026

COST OF INACTION - THE MATH

8 Cameras x 12 Operating Hours/Day x 6 Operating Days/Week = 576 Video-Hours Generated Weekly

The Bottleneck: Yet, the safety team can realistically review only **4 hours per week manually**.

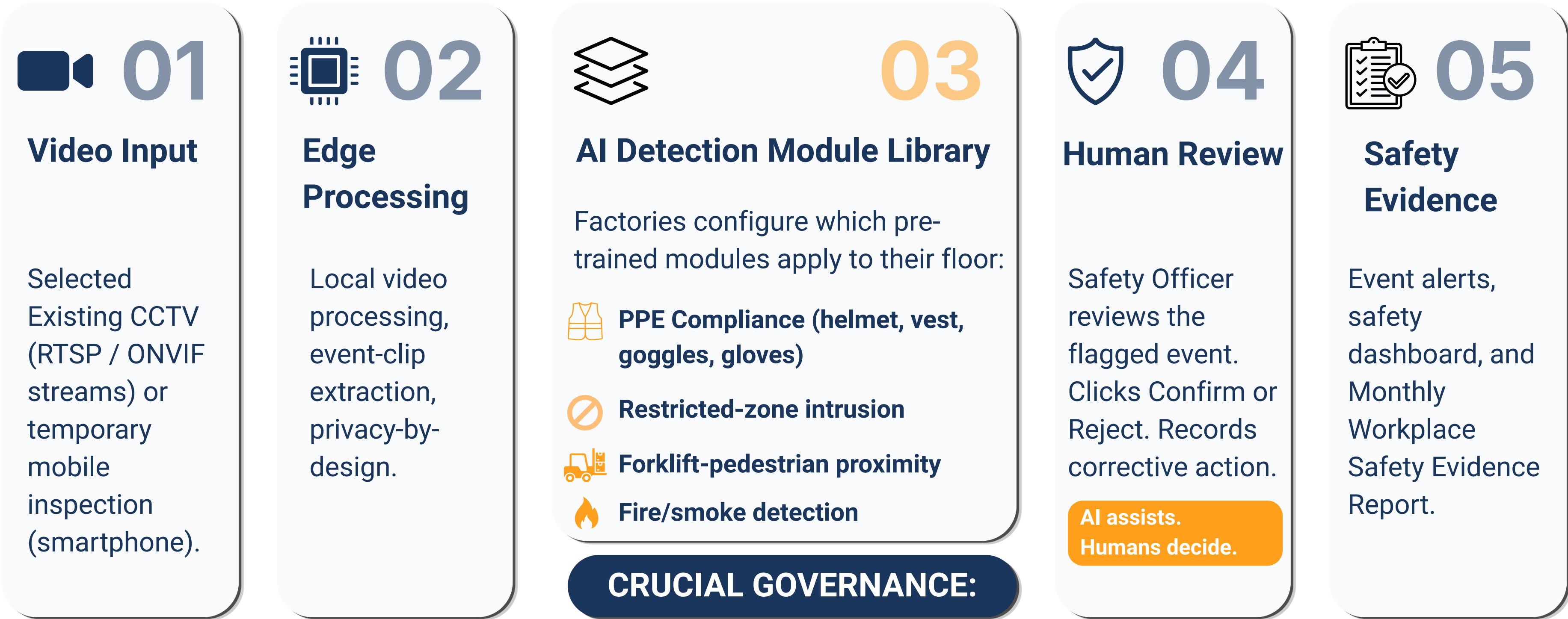
The Deficit: This leaves **572 hours of unmonitored risk** every single week.



Conclusion : The vast majority of preventable risks go undocumented, leading to repeated hazards, fragmented safety records, and costly downtime.

Product Architecture :

From Existing Cameras to Safety Evidence



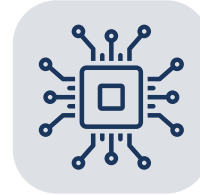
No facial recognition | No fatigue inference | No automatic disciplinary action | No complete labor-compliance certification

Legacy-Compatible, Privacy-Preserving Architecture



01 Data Collection Layer

RTSP/ONVIF-compatible camera streams (subject to site assessment). Temporary mobile mode for blind spots. No expensive AR/VR equipment required.



02 Edge Processing Layer

On-site edge device. Reduces cloud bandwidth. Buffers locally during unstable internet connections.



03 Output Layer

Cloud-hosted event dashboard and factory-authorized report export.

Privacy Safeguards :

- ✓ No facial recognition
- ✓ No fatigue or productivity scoring
- ✓ Human review before action
- ✓ Role-based access

Strategic Alignment: The Four-Pillar Impact Matrix

Worker protection is the core value created by KantaForge, not a side CSR programme.

PILLAR	PRODUCT FUNCTION	MEASURABLE OUTCOME
 ESG (Social)	Converts observable safety events into human-reviewed records.	Confirmed safety events, resolution rate.
 SDG 8 (Decent Work)	Flags predefined workplace risks.	Workers covered, reduction in repeated risks.
 SDG 9 (Industry Innovation)	Reuses existing CCTV to digitalise legacy workflows.	SMEs onboarded, monitored zones digitised.
 NIMP 2030	Supports affordable technology adoption among SMEs.	Legacy systems integrated, pilot-to-paid conversion.
 Malaysian Social Issue	Blue-collar / B40 worker safety monitoring.	Corrective actions completed.

Ethical AI: **Protecting Workers**, Not Policing Them



CORE PRINCIPLE

KantaForge detects unsafe conditions and system hazards, not "bad workers".

Privacy-by-Design Features:



Edge-processing by default (no raw video stockpiling).



Worker notification protocols.



Role-based access control.



Human-in-the-loop :

The AI cannot punish. Every generated alert requires a human Safety Officer's confirmation before it becomes an official record.

Platform SaaS Model & Go-To-Market Strategy

 **Target Customer:** Malaysian Manufacturing SMEs (CNC, precision metal).



Pricing Structure



One-time Setup Fee

Covers CCTV assessment, zone mapping, and edge device configuration.



Monthly Platform Subscription

Base platform access + configurable AI module add-ons (PPE, proximity, etc.).



Go-To-Market Funnel

~ **Start Small, Expand After Validation**

1

One-camera, one-zone pilot.

2

Convert pilot into a paid subscription.

3

Cross-sell additional AI modules & Multi-Zone Expansion.

4

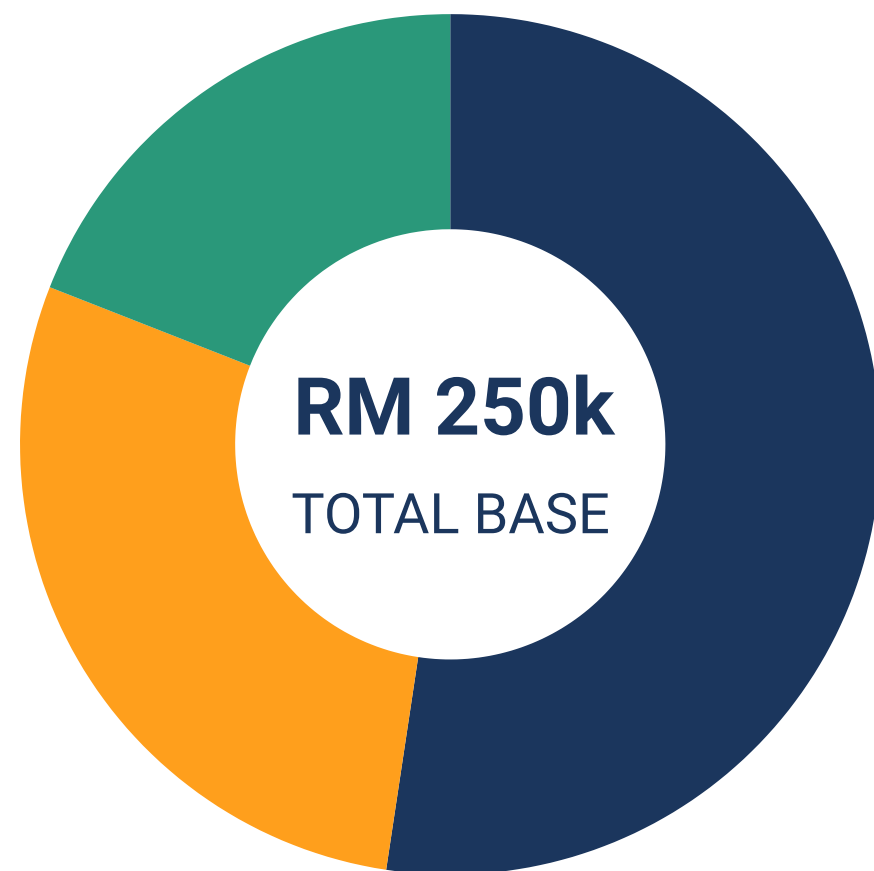
Scale through safety consultants and CCTV partners.

Capital Allocation & External Grant Validation



Capital Allocation Breakdown

RM 250,000 Base Allocation
Hard Cap - No further injection



■ MVP build

Spring Boot + Computer Vision module library

■ Edge devices & cloud

Infrastructure setup for Year 1

■ Pilot deployments & compliance

First pilot deployments, legal / privacy compliance, and onboarding



External Grant Validation

SELECTED GRANT

Cradle CIP Sprint

Provides up to RM 600,000; we cap at RM 250,000



Eligibility Checklist



Registered as Sdn Bhd (To be completed pre-application)



Company under 7 years old



At least 51% Malaysian ownership

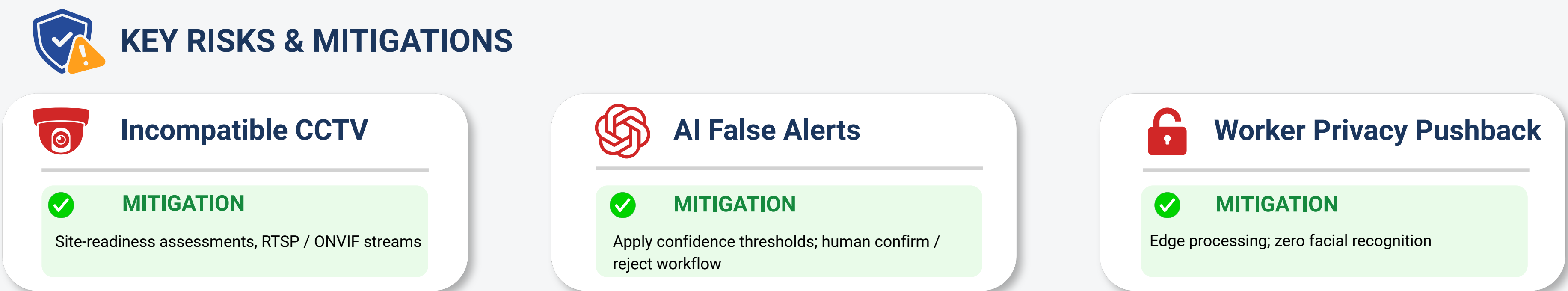
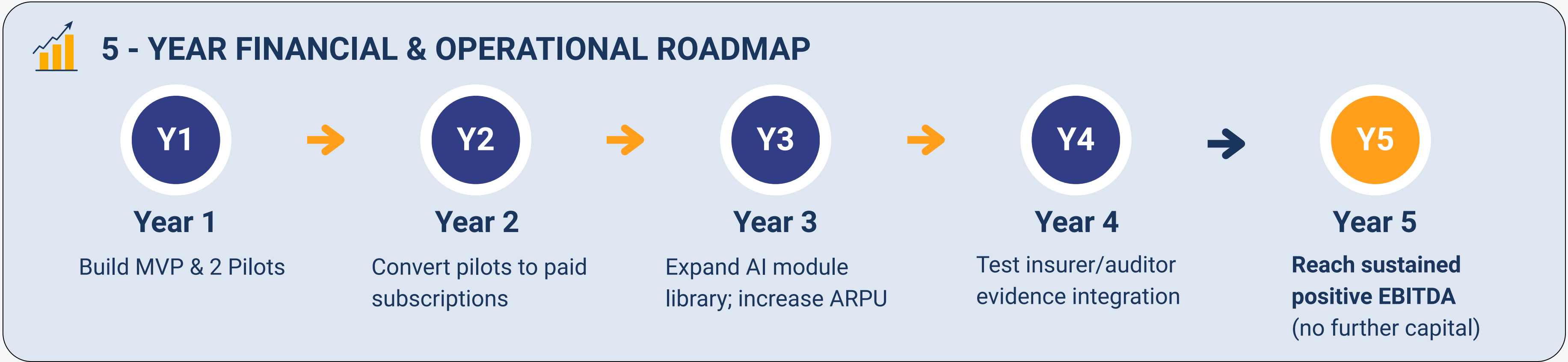


Cumulative revenue below RM 5,000,000



≥60% of grant funds allocated to commercialization

Path to EBITDA Breakeven & Risk Mitigation



Appendix 1 : Primary Research Evidence



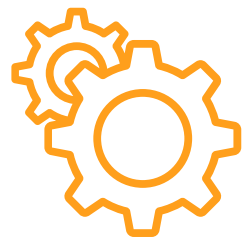
Voice of Customer (VoC) +
Field Interview Insights

INTERVIEWEE NAME	ROLE	FACTORY TYPE	DATE	EXACT USABLE QUOTATION
Mr. Tan	EHS Manager	Manufacturing Company	Aug 5, 2026	“CCTV is mostly reviewed after an incident. Earlier safety alerts would help us respond faster.”
Mr. Lee	Lead ISO 45001 Author	Industrial Safety Company	Aug 6, 2026	“Detection alone is not enough. Auditors need evidence of review, corrective action, and closure.”
Mr. Khor	Head of Maintenance	Metal Stamping	Aug 7, 2026	“We still rely a lot on manual monitoring. If the system can detect safety issues earlier and alert us immediately, it would help us respond much faster.”

Appendix 2 : Financial Assumptions



All uncertain figures are illustrative assumptions pending pilot validation.



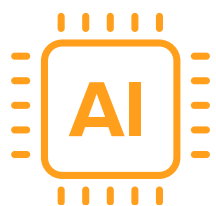
Setup Fee

RM 4500



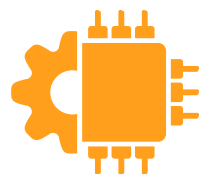
Base Platform Subscription

RM 800
/month



Additional AI Module

RM 250
/month

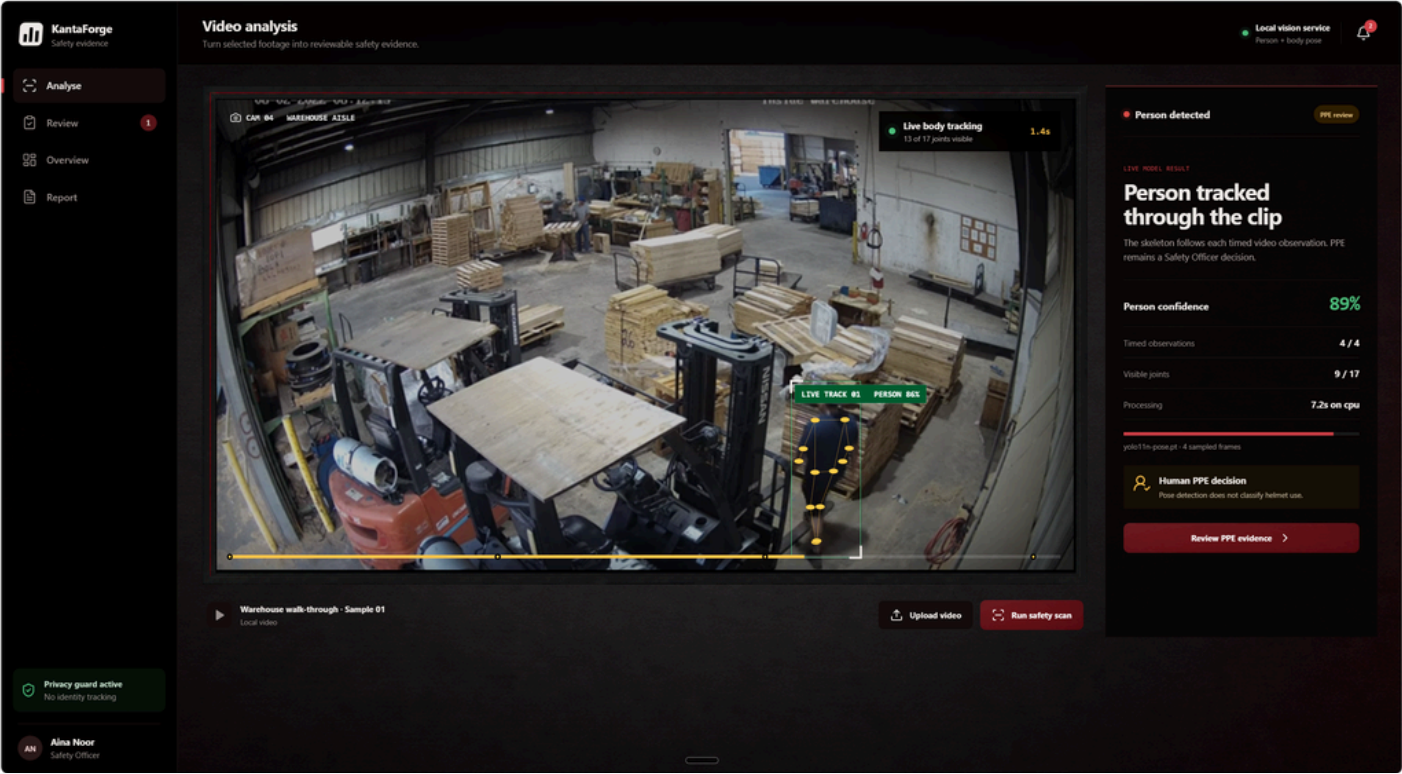


Edge Hardware Cost

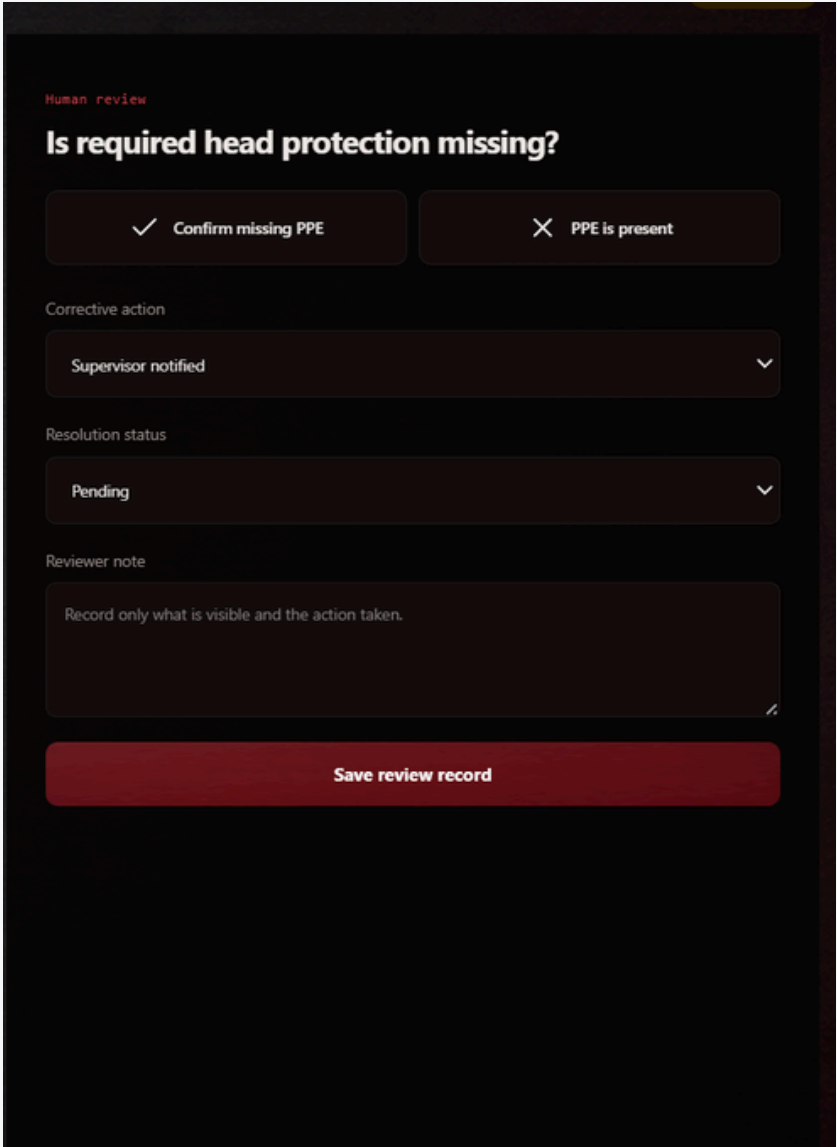
RM 3500



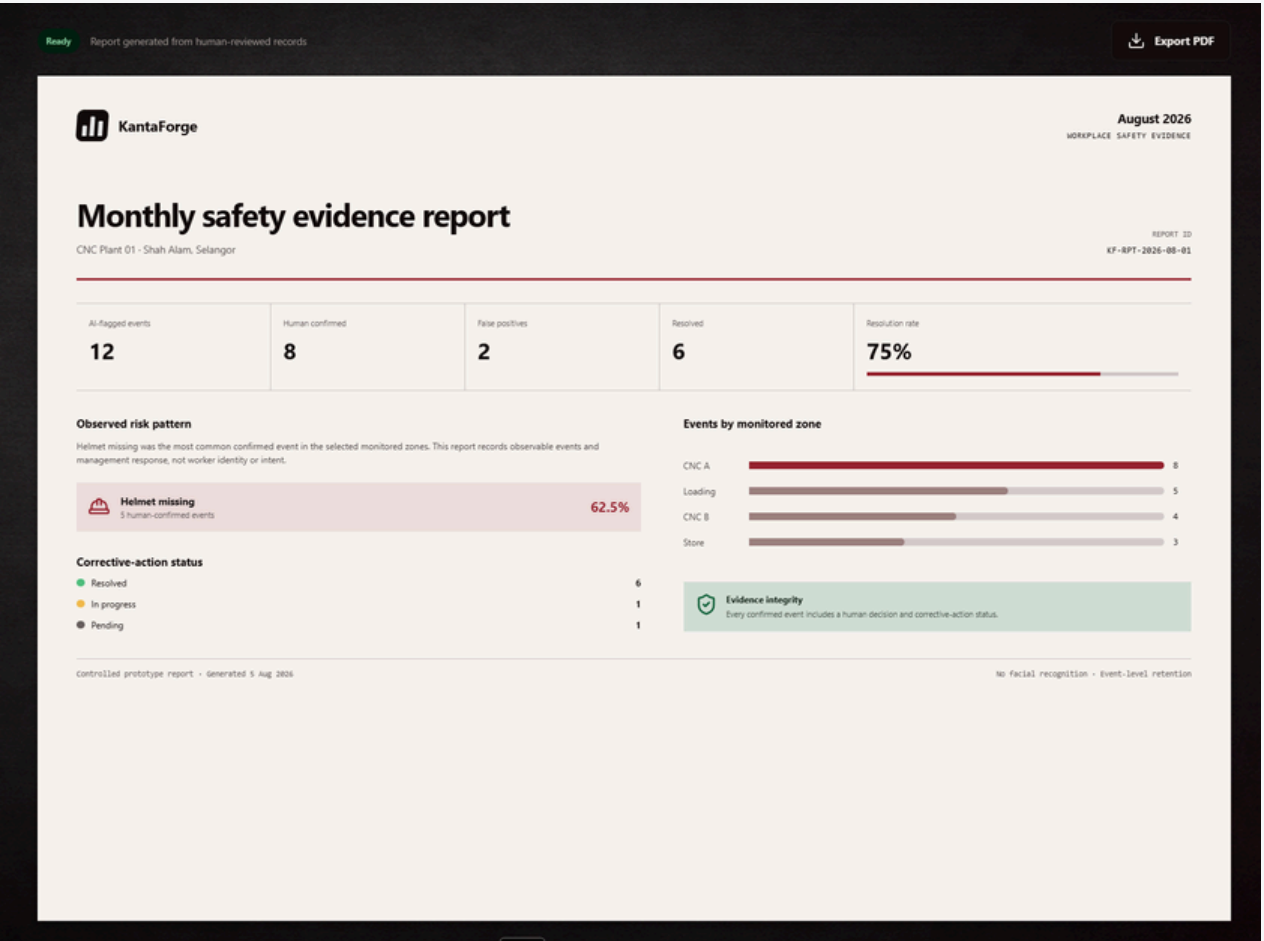
Appendix 3 : Prototype & Technical Evidence



CCTV Analysis



Human Review & Corrective action



Monthly Safety Evidence Report

Github prototype link: <https://github.com/25006675-png/kantaforge.git>

References

1. Department of Statistics Malaysia (DOSM) Injury Data

Department of Statistics Malaysia. (2024, November 12). National occupational injury and disease statistics 2023. <https://www.dosm.gov.my>

2. Cradle CIP Sprint Grant Guidelines

Cradle Fund. (n.d.). CIP Sprint: Commercialisation grant for tech start-ups. Retrieved August 5, 2026, from <https://cradle.com.my/our-funding/cip-sprint/>

3. ONVIF Video-Streaming Standard

ONVIF. (n.d.). Profile T: Advanced Video Streaming for IP-Based Video Systems. <https://www.onvif.org/profiles/profile-t/>